

Umair Ahmed

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ABOUT ME

Multidisciplinary Computer Science graduate experienced in full-stack development (**Python, Java, TypeScript**) and applied machine learning. Skilled in database design (**PostgreSQL, MySQL**) and deploying custom software through DevOps practices, including **CI/CD** pipelines and containerized environments using **Docker**.

EDUCATION

Bachelor of Science in Computer Science

Aston University

2022-Oct 2025

Focus: Software Engineering, Databases, Distributed Systems, OOP, AI/ML

Skills: Python, Java, SQL, Linux, Git, HTML & CSS

PROJECTS

Reinforcement Learning Environment (Python, 2025)

- Developed a custom Gymnasium-compatible RL environment simulating a robotic arm flipping an omelette, with realistic physics and reward shaping in PyBullet.
- Implemented training using PPO with iterative feedback tuning, testing via simulation GUI, and debugging for system reliability and performance.

Shorty – URL Shortener Web App (Java, Spring Boot, Docker, 2025)

- Built a production-ready URL shortener with REST APIs, PostgreSQL integration, and environment-based configuration in Java Spring Boot.
- Containerized and deployed via Docker and Caddy reverse proxy on a Hetzner VPS, implementing CI/CD automation for smooth and scalable updates.

DocuQA – RAG Chatbot (Python, FastAPI, Next.js, ChromaDB, 2025)

- Developed an intelligent document-query system allowing users to upload PDFs or Markdown files and interact with them through a chat interface powered by retrieval-augmented generation (RAG).
- Built a FastAPI backend with ChromaDB vector storage and integrated a Next.js/TypeScript frontend for real-time responses and clean UI.

SKILLS

Programming & OOP: Java (+ Spring Boot), Python, .NET, Lua, JS, TS, React, Tailwind, OOP & DSA

Databases: PostgreSQL, MySQL, ChromaDB

Tools: Docker, Caddy, Git, VS Code, IntelliJ, Visual Studio

Additional Knowledge: Containerisation, CI/CD automation, Cloud deployment (Hetzner VPS), Reverse Proxy & TLS, REST APIs, QA & testing, TDD (Test Driven Development)

Personal Skills: Leadership, Problem-solving, Adaptability, Communication, Teamwork

INTERESTS

Career Interests: Full-stack development, DevOps, Machine Learning, Databases

Personal Interests: Working out, Hiking, Gaming, Basketball, Gardening, Cooking